

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 October 2025 Worked Solutions

Oct 2025 paper rebuilt with the worked solution after each question.

10

paper questions

WMA11

Pure 1

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - eliteigcse.com

P1

PAST PAPER

WMA11/01 October 2025

Oct 2025 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Differentiation

1. The curve C has equation

$$y = 6x^2 + 3\sqrt{x} + \frac{5}{8} \quad x > 0$$

- (a) Find $\frac{dy}{dx}$ writing the answer in simplest form.

(3)

The point $P\left(\frac{1}{4}, \frac{5}{2}\right)$ lies on C .

- (b) Find the equation of the tangent to C at P , writing your answer in the form $y = mx + c$ where m and c are integers.

(3)

(Total for Question 1 is 6 marks)

Worked Solution - Question 1

1. Rewrite the curve

$$y = 6x^2 + 3x^{1/2} + \frac{5}{8}.$$

2. Differentiate

$$\frac{dy}{dx} = 12x + \frac{3}{2}x^{-1/2} = 12x + \frac{3}{2\sqrt{x}}.$$

3. Find the tangent gradient

$$\text{At } x = \frac{1}{4}, \frac{dy}{dx} = 12\left(\frac{1}{4}\right) + \frac{3}{2}\left(\frac{1}{4}\right)^{-1/2} = 3 + 3 = 6.$$

4. Use point P

$$\text{The tangent passes through } P\left(\frac{1}{4}, \frac{5}{2}\right), \text{ so } y - \frac{5}{2} = 6\left(x - \frac{1}{4}\right).$$

5. Simplify

$$y - \frac{5}{2} = 6x - \frac{3}{2}, \text{ hence } y = 6x + 1.$$

Final answer

$$(a) \frac{dy}{dx} = 12x + \frac{3}{2\sqrt{x}}.$$

$$(b) y = 6x + 1.$$

Question 2

Integration

2. Find

$$\int (x-3)^2 (2x+5) \, dx$$

writing the answer in simplest form.

(5)

(Total for Question 2 is 5 marks)

Worked Solution - Question 2

1. Expand the square

$$(x - 3)^2 = x^2 - 6x + 9.$$

2. Multiply out

$$(x^2 - 6x + 9)(2x + 5) = 2x^3 + 5x^2 - 12x^2 - 30x + 18x + 45.$$

3. Collect terms

$$(x - 3)^2(2x + 5) = 2x^3 - 7x^2 - 12x + 45.$$

4. Integrate term by term

$$\int(2x^3 - 7x^2 - 12x + 45)dx = \frac{1}{2}x^4 - \frac{7}{3}x^3 - 6x^2 + 45x + c.$$

Final answer

$$\frac{1}{2}x^4 - \frac{7}{3}x^3 - 6x^2 + 45x + c.$$

Question 3

Basic Trigonometry

3. The sides of a triangle are in the ratio 2:4:5

(a) Find the size of the **largest** angle of the triangle, in degrees, to one decimal place.

(2)

Given that the area of the triangle is 140 cm^2

(b) find the length of the **shortest** side, in cm, to one decimal place.

(3)

(Total for Question 3 is 5 marks)

Worked Solution - Question 3

Topic group

1. Use the side ratio

Take the side lengths as $2x$, $4x$ and $5x$. The largest angle is opposite the side $5x$.

2. Use the cosine rule

$$(5x)^2 = (2x)^2 + (4x)^2 - 2(2x)(4x) \cos \theta.$$

3. Find the angle

$$25 = 4 + 16 - 16 \cos \theta, \text{ so } \cos \theta = -\frac{5}{16} \text{ and } \theta = 108.2^\circ \text{ to one decimal place.}$$

4. Use the area formula

$$140 = \frac{1}{2}(2x)(4x) \sin 108.2^\circ = 4x^2 \sin 108.2^\circ.$$

5. Find the scale factor

$$x^2 = \frac{140}{4 \sin 108.2^\circ}, \text{ so } x \approx 6.070.$$

6. Shortest side

The shortest side is $2x \approx 12.1$ cm.

Final answer

(a) 108.2° .

(b) 12.1 cm.

Question 4**Indices & Surds**

4. **In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

(i) Using the laws of indices, solve

$$3^{2p-5} = \frac{1}{9}\sqrt{27}$$

(3)

(ii) Find the real roots of the equation

$$x^2 + 1 = \frac{36}{x^2 - 4}$$

(4)

(Total for Question 4 is 7 marks)

Worked Solution - Question 4

1. Rewrite the right-hand side

$$\frac{1}{9}\sqrt{27} = 3^{-2} \cdot 3^{3/2} = 3^{-1/2}.$$

2. Equate powers of 3

$$3^{2p-5} = 3^{-1/2}, \text{ so } 2p - 5 = -\frac{1}{2}.$$

3. Solve for p

$$2p = \frac{9}{2}, \text{ hence } p = \frac{9}{4}.$$

4. Clear the denominator

$$x^2 + 1 = \frac{36}{x^2 - 4} \text{ gives } (x^2 + 1)(x^2 - 4) = 36.$$

5. Form a quadratic in x squared

$$x^4 - 3x^2 - 4 = 36, \text{ so } x^4 - 3x^2 - 40 = 0.$$

6. Factor

$$(x^2 - 8)(x^2 + 5) = 0.$$

7. Use real roots

$$x^2 = 8, \text{ so } x = \pm 2\sqrt{2}. \text{ The equation } x^2 = -5 \text{ gives no real roots.}$$

Final answer

$$(i) p = \frac{9}{4}. \quad (ii) x = \pm 2\sqrt{2}.$$

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10 marks

Question 5

Quadratics

5.

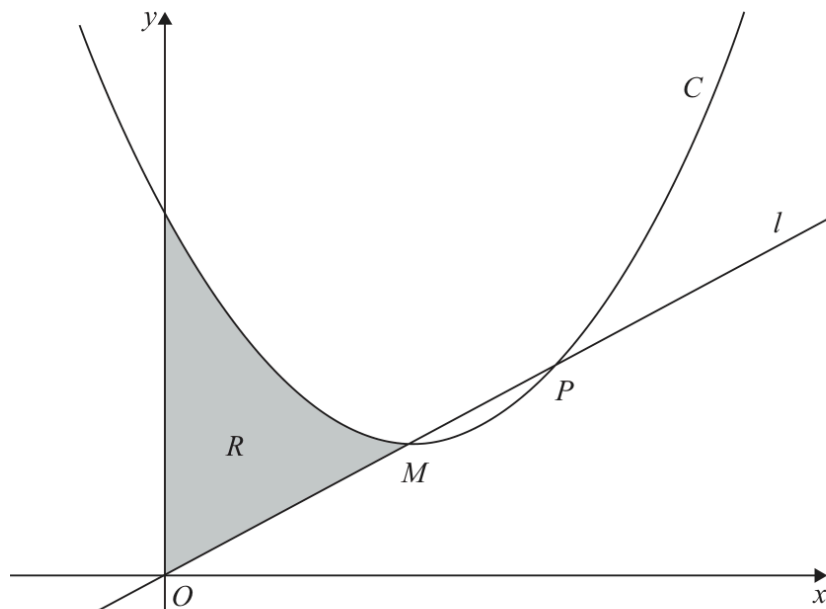


Figure 1

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

- (a) Express $2x^2 - 16x + 50$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

Figure 1 shows a sketch of the curve C with equation $y = 2x^2 - 16x + 50$

Given that in Figure 1

- M is the minimum point on C
- line l passes through the origin and intersects C at the points M and P

- (b) find, using algebra and showing your working, the x coordinate of P .

(5)

The region R is shown shaded in Figure 1.

- (c) Use inequalities to fully define R .

(2)

(Total for Question 5 is 10 marks)

Worked Solution - Question 5

1. Complete the square

$$2x^2 - 16x + 50 = 2(x^2 - 8x) + 50 = 2[(x - 4)^2 - 16] + 50.$$

2. Simplify the completed square

$$2x^2 - 16x + 50 = 2(x - 4)^2 + 18.$$

3. Find M

The minimum point is $M = (4, 18)$.

4. Find the line through O and M

Since l passes through $(0, 0)$ and $(4, 18)$, its gradient is $\frac{18}{4} = \frac{9}{2}$, so $l : y = \frac{9}{2}x$.

5. Find the second intersection

$$\text{Set } 2x^2 - 16x + 50 = \frac{9}{2}x.$$

6. Form the quadratic

Multiplying by 2 gives $4x^2 - 32x + 100 = 9x$, so $4x^2 - 41x + 100 = 0$.

7. Factor and select P

$$4x^2 - 41x + 100 = (x - 4)(4x - 25) = 0. \text{ The point } M \text{ has } x = 4, \text{ so } P \text{ has } x = \frac{25}{4}.$$

8. Define the region

The shaded region is above the line, below the curve, and between the y-axis and M .

9. Write inequalities

$$\frac{9}{2}x \leq y \leq 2x^2 - 16x + 50 \text{ with } 0 \leq x \leq 4.$$

Final answer

$$(a) 2(x - 4)^2 + 18.$$

$$(b) x_P = \frac{25}{4}.$$

$$(c) \frac{9}{2}x \leq y \leq 2x^2 - 16x + 50, 0 \leq x \leq 4.$$

Question 6

Graphs of Functions

6. (a) Sketch the graph of the curve C with equation

$$y = \frac{4k}{x - 2k}$$

where k is a positive constant.

On your sketch show

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

The straight line l has equation

$$y = 6 - 2x$$

Given that there is at least one point of intersection between l and C ,

- (b) find the range of possible values of k .

(5)

(Total for Question 6 is 9 marks)

Worked Solution - Question 6

1. Find coordinate-axis intersections

$y = \frac{4k}{x - 2k}$ has no x-intercept because the numerator is non-zero. At $x = 0$,
 $y = \frac{4k}{-2k} = -2$, so the y-intercept is $(0, -2)$.

2. Find the vertical asymptote

The denominator is zero when $x = 2k$, so the vertical asymptote is $x = 2k$.

3. Sketch shape

The curve has one branch in quadrant I to the right of $x = 2k$, and one branch passing through $(0, -2)$ and approaching the x-axis from below as $x \rightarrow -\infty$.

4. Set line and curve equal

$$\frac{4k}{x - 2k} = 6 - 2x.$$

5. Form a quadratic in x

$$4k = (6 - 2x)(x - 2k) = 6x - 12k - 2x^2 + 4kx, \text{ so}$$
$$2x^2 - (4k + 6)x + 16k = 0.$$

6. Use at least one intersection

For at least one real intersection, the discriminant must satisfy $b^2 - 4ac \geq 0$.

7. Find the discriminant

$$(4k + 6)^2 - 4(2)(16k) \geq 0, \text{ so } 16k^2 - 80k + 36 \geq 0.$$

8. Solve in k

$$4k^2 - 20k + 9 \geq 0, \text{ so } (2k - 1)(2k - 9) \geq 0.$$

9. Use k positive

Therefore $k \leq \frac{1}{2}$ or $k \geq \frac{9}{2}$. Since k is positive, \$0

Final answer

(a) y-intercept $(0, -2)$, vertical asymptote $x = 2k$.

(b) 0

Question 7

Radians

7.

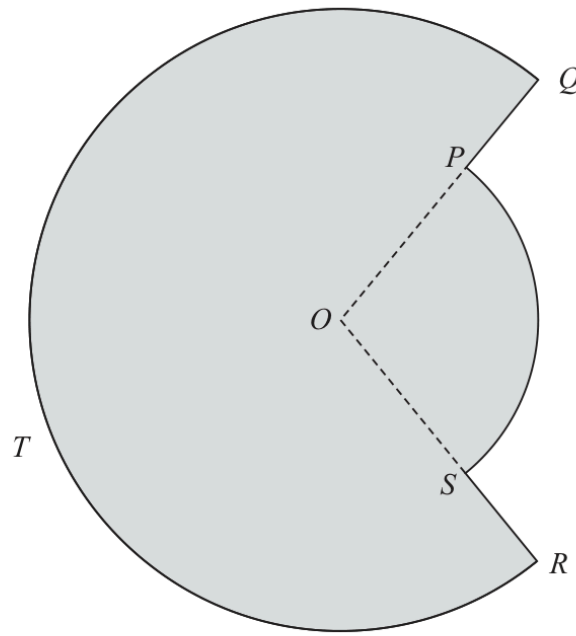


Figure 2

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 2 shows the plan view of a design for a swimming pool.

The design consists of a sector POS of a circle centre O joined to a major sector $QORTQ$ of a different circle, also with centre O .

Given that

- angle POS is 1.65 radians
- the area of sector POS is 30 m^2
- $PQ = 2.8 \text{ m}$

- (a) show that, to 3 significant figures, $OQ = 8.83 \text{ m}$ (3)
- (b) Find the total surface area of the swimming pool in m^2 to the nearest integer. (3)
- (c) Find the total perimeter of the swimming pool in metres to 2 significant figures. (3)

(Total for Question 7 is 9 marks)

Worked Solution - Question 7

1. Find OP from the small sector

Let $OP = r$. The area of sector POS is 30, so $\frac{1}{2}r^2(1.65) = 30$.

2. Calculate OP

$$r^2 = \frac{60}{1.65}, \text{ so } r = 6.030\dots \text{ m.}$$

3. Show OQ

$OQ = OP + PQ = 6.030\dots + 2.8 = 8.830\dots$, so $OQ = 8.83$ m to 3 significant figures.

4. Major sector angle

The major sector $QORTQ$ has angle $2\pi - 1.65$ radians.

5. Total area

$$\text{Area} = 30 + \frac{1}{2}(8.830\dots)^2(2\pi - 1.65) = 210.6\dots$$

6. Round area

The total surface area is **211 m²** to the nearest integer.

7. Arc lengths

The inner arc PS has length $6.030\dots(1.65) = 9.95\dots$, and the outer major arc has length $8.830\dots(2\pi - 1.65) = 40.91\dots$

8. Add straight sides

The straight sides PQ and SR each have length 2.8 m.

9. Perimeter

$9.95\dots + 40.91\dots + 2.8 + 2.8 = 56.46\dots$, so the perimeter is **56 m** to 2 significant figures.

Final answer

(a) $OQ = 8.83$ m.

(b) 211 m².

(c) **56 m**.

Question 8

Integration

8.

In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

A curve has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = 2x + \frac{8}{x^2} + k$, where k is a constant
- the equation of the tangent to the curve at $x = \sqrt{2}$ is $y = 5x - 3\sqrt{2}$

(a) find the exact value of k .

(2)

(b) Find an equation of the normal to the curve at $x = \sqrt{2}$

(2)

(c) Find $f(x)$, writing your answer in simplest form.

(4)

(Total for Question 8 is 8 marks)

Worked Solution - Question 8

1. Use tangent gradient

The tangent has equation $y = 5x - 3\sqrt{2}$, so the gradient at $x = \sqrt{2}$ is 5.

2. Find k

$$f'(\sqrt{2}) = 2\sqrt{2} + \frac{8}{(\sqrt{2})^2} + k = 2\sqrt{2} + 4 + k = 5, \text{ so } k = 1 - 2\sqrt{2}.$$

3. Find the point on the curve

At $x = \sqrt{2}$, the tangent gives $y = 5\sqrt{2} - 3\sqrt{2} = 2\sqrt{2}$.

4. Normal gradient

The tangent gradient is 5, so the normal gradient is $-\frac{1}{5}$.

5. Equation of the normal

$$y - 2\sqrt{2} = -\frac{1}{5}(x - \sqrt{2}), \text{ hence } y = -\frac{1}{5}x + \frac{11\sqrt{2}}{5}.$$

6. Integrate f prime

$$f(x) = \int \left(2x + \frac{8}{x^2} + k \right) dx = x^2 - \frac{8}{x} + kx + c.$$

7. Substitute k

$$f(x) = x^2 - \frac{8}{x} + (1 - 2\sqrt{2})x + c.$$

8. Use the point

$$2\sqrt{2} = 2 - \frac{8}{\sqrt{2}} + (1 - 2\sqrt{2})\sqrt{2} + c = -2 - 3\sqrt{2} + c.$$

9. Find c

$$c = 2 + 5\sqrt{2}, \text{ so } f(x) = x^2 - \frac{8}{x} + (1 - 2\sqrt{2})x + 2 + 5\sqrt{2}.$$

Final answer

(a) $k = 1 - 2\sqrt{2}$.

(b) $y = -\frac{1}{5}x + \frac{11\sqrt{2}}{5}$.

(c) $f(x) = x^2 - \frac{8}{x} + (1 - 2\sqrt{2})x + 2 + 5\sqrt{2}$.

Question 9

Transformations

9.

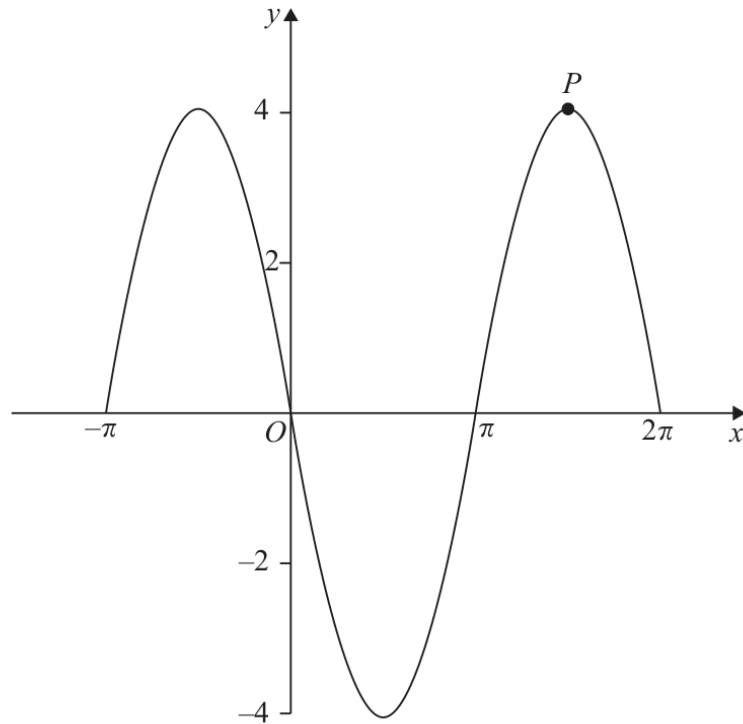


Figure 3

Figure 3 shows a sketch of part of the graph of the trigonometric function with equation $y = f(x)$

(a) Write down an expression for $f(x)$

(2)

The point P lies on $y = f(x)$ and is shown in Figure 3.

(b) State the coordinates of the point to which P is transformed when the graph of $y = f(x)$ is transformed to the graph with equation

(i) $y = f\left(x - \frac{\pi}{6}\right)$

(2)

(ii) $y = -\frac{1}{2}f(x)$

(2)

(Total for Question 9 is 6 marks)

Worked Solution - Question 9

1. Identify amplitude and direction

The graph has amplitude 4, crosses the origin, and initially moves downwards, so $f(x) = -4 \sin x$.

2. Identify P

The labelled maximum is at $P = \left(\frac{3\pi}{2}, 4\right)$.

3. Transform y equals f of x minus pi over 6

$y = f\left(x - \frac{\pi}{6}\right)$ translates the graph right by $\frac{\pi}{6}$.

4. New coordinates for part b(i)

P becomes $\left(\frac{3\pi}{2} + \frac{\pi}{6}, 4\right) = \left(\frac{5\pi}{3}, 4\right)$.

5. Transform y equals negative half f(x)

$y = -\frac{1}{2}f(x)$ leaves x-coordinates unchanged and multiplies y-coordinates by $-\frac{1}{2}$.

6. New coordinates for part b(ii)

P becomes $\left(\frac{3\pi}{2}, -2\right)$.

Final answer

$$(a) f(x) = -4 \sin x.$$

$$(b)(i) \left(\frac{5\pi}{3}, 4 \right).$$

$$(b)(ii) \left(\frac{3\pi}{2}, -2 \right).$$

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10 marks

Question 10

Differentiation

10.

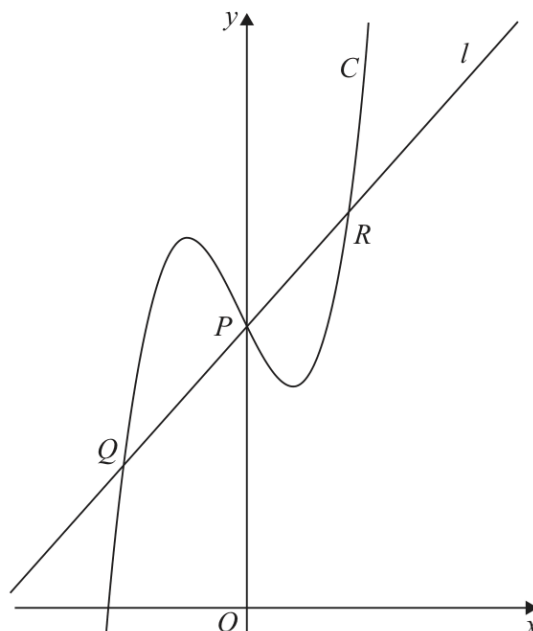


Figure 4

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 4 shows a sketch of the curve C with equation

$$y = 2x^3 + \frac{1}{2}x^2 - 2x + 5$$

The line l is the normal to C at the point P where $x = 0$

The line l also intersects C at points Q and R as shown in Figure 4.

(a) Find, using algebra, the x coordinate of point Q .

(6)

The point T lies on C .

Given that

- the tangent to C at T is parallel to l
- the x coordinate of T is positive

(b) find, using algebra, the exact x coordinate of T .

(4)

(Total for Question 10 is 10 marks)

Worked Solution - Question 10

1. Differentiate C

$$y = 2x^3 + \frac{1}{2}x^2 - 2x + 5, \text{ so } \frac{dy}{dx} = 6x^2 + x - 2.$$

2. Find the tangent gradient at P

At P , $x = 0$, so the tangent gradient is -2 .

3. Equation of the normal

The normal gradient is $\frac{1}{2}$ and $P = (0, 5)$, so $l : y = \frac{1}{2}x + 5$.

4. Find intersections with the normal

$$\text{Set } 2x^3 + \frac{1}{2}x^2 - 2x + 5 = \frac{1}{2}x + 5.$$

5. Form and factor the cubic

$$2x^3 + \frac{1}{2}x^2 - \frac{5}{2}x = 0. \text{ Multiplying by 2 gives}$$
$$4x^3 + x^2 - 5x = 0 = x(4x^2 + x - 5).$$

6. Solve for x

$$x(4x + 5)(x - 1) = 0, \text{ so the intersections have x-coordinates } 0, -\frac{5}{4} \text{ and } 1.$$

7. Select point Q

$$\text{The point } Q \text{ is the left-hand intersection, so } x_Q = -\frac{5}{4}.$$

8. Parallel tangent condition

$$\text{The tangent at } T \text{ is parallel to } l, \text{ so its gradient is } \frac{1}{2}.$$

9. Set derivative equal to one half

$$6x^2 + x - 2 = \frac{1}{2}, \text{ so } 12x^2 + 2x - 5 = 0.$$

10. Use the positive solution

$$x = \frac{-2 \pm \sqrt{4 + 240}}{24} = \frac{-1 \pm \sqrt{61}}{12}. \text{ Since } x_T \text{ is positive, } x_T = \frac{-1 + \sqrt{61}}{12}.$$

Final answer

$$(a) x_Q = -\frac{5}{4}.$$

$$(b) x_T = \frac{-1 + \sqrt{61}}{12}.$$